📄 Project Report: Library Pro Enterprise v5.0

1. Abstract

The Library Pro Enterprise system is a robust, Python-based desktop application designed to streamline the management of library resources. It utilizes a secure Role-Based Access Control (RBAC) system to distinguish between Administrative and Student functionalities. The core innovation of version 5.0 is the Admin Verification Desk, which ensures that book returns are physically verified by staff before the digital record is updated, maintaining 100% data integrity.

2. System Requirements

Operating System: Windows 10/11, macOS, or Linux.

Programming Language: Python 3.10 or higher.

Database: SQLite3 (Serverless Relational Database).

Libraries: Tkinter (GUI), SQLite3 (Data), Pillow (Image processing).

3. Proposed System Design (Flowchart)

The system follows a linear security logic for transactions:

Authentication: User logs in as Admin or Student.

Discovery: Student searches the inventory via a real-time filter.

Transaction: Student issues a book (Status changes to Issued).

Return Request: Student returns the book (Status changes to Pending Return).

Verification: Admin approves the return in the Verification Desk (Status changes to Available).

4. Database Schema (Backend)

The system operates on three relational tables:

Users Table: Stores username, password, and role (Admin/Student).

Books Table: Tracks book_id, title, author, and status.

Transactions Table: Records issue_date, due_date, and verified_status.

6. Conclusion

This project successfully automates the manual labor of library tracking. Through the development of Library Pro v5.0, I have gained deep experience in Python GUI development, relational database management, and the implementation of secure business logic workflows.